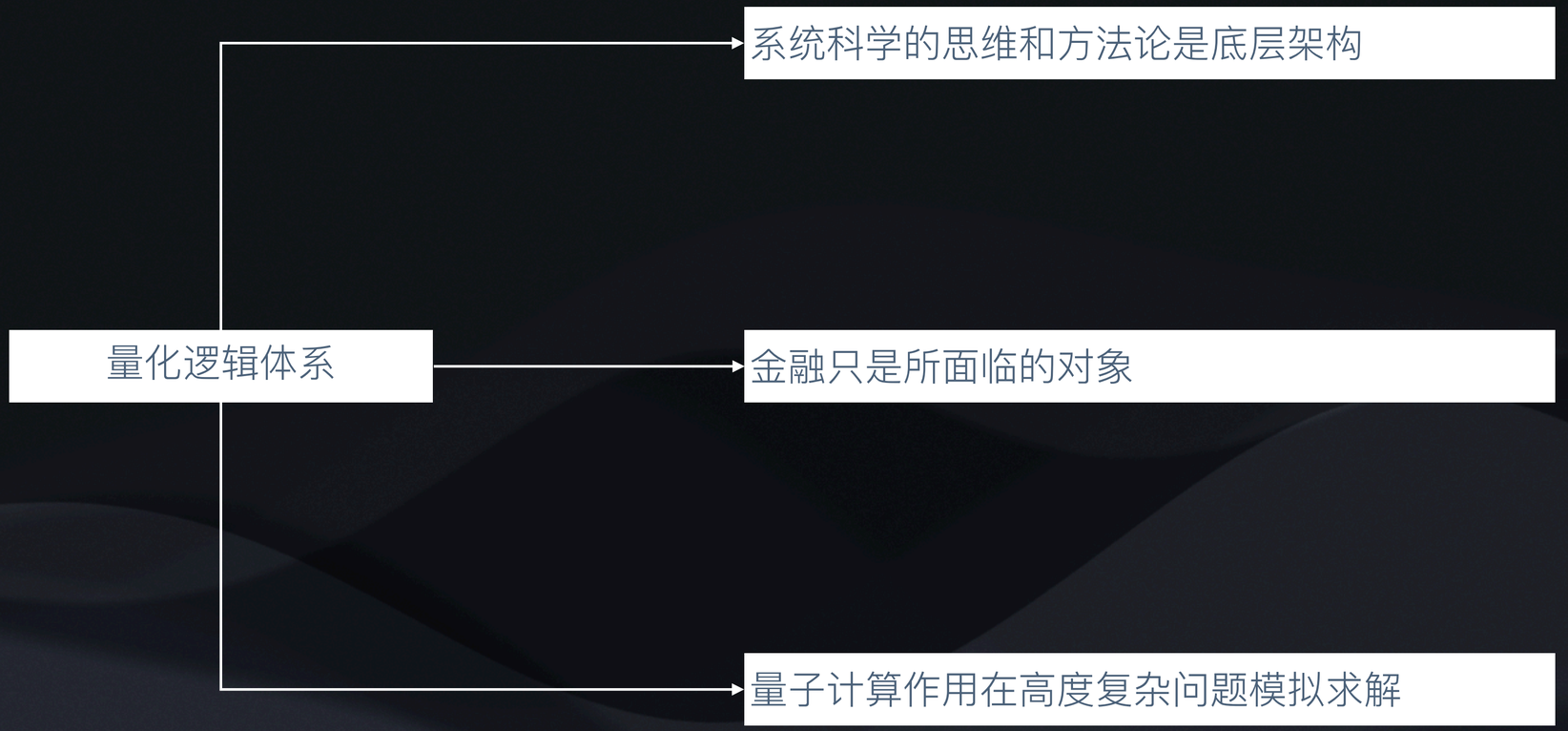
量子计算机是一项系统工程

王浪涛

整体框架

量化金融核心在量化



系统方法论的哲学基础

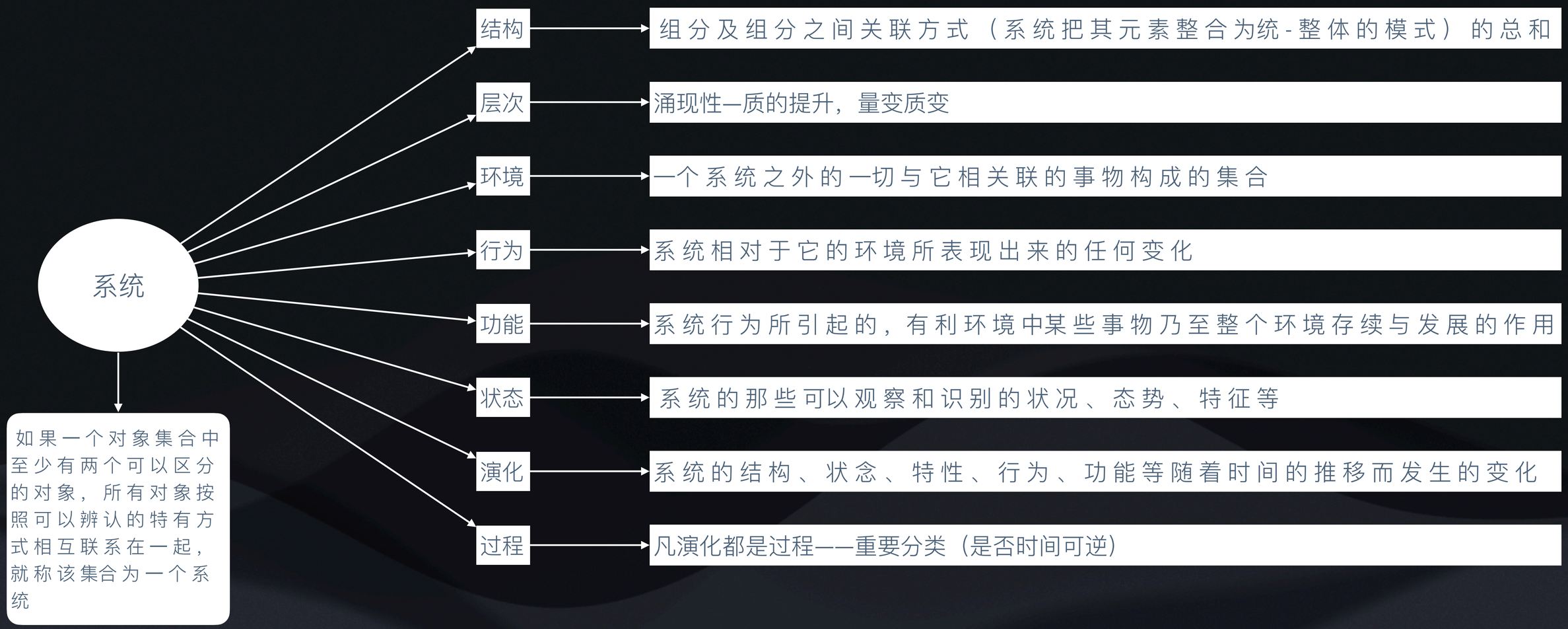
辩证法

- 系统方法论的哲学依据,归根到底是唯物辩证法。某些西方系统科学家不愿公开承认这一点，但他们的工作成就实质上都得益于辨证法。多数系统科学大家明确承认辨证法对系统研究的指导作用。贝塔朗菲承认马克思的辨证法对今天被称为一般系统论的理论观念的发展作出了自己的贡献。丘奇曼预言未来的系统分析必定会提出一种新的哲学，其"主旨概念将是辨证的学习过程"。普利高津主张，"我们需要一种更加辨证的自然观"。哈肯在谈到协同学的"哲学方面"时，明确应用了对立统一、量变质变等辩证法规律。钱学森更是不进余力地宣传系统科学必领以马克恐主义哲学为指导，自觉地应用辩证法于系统研究。几十年来,中国系统科学界始终坚持以唯物辨证法来指导系统研究和教学。

- 辩证法的核心是对立统一。用之于系统研究,就是强调还原论方法和整体论方法的结合，分析方法与综合方法的结合，定性描述与定量描述的结合,局部描达与整体描述的结合，确定性描述与不确定性描述的结合，静力学描述与动力学描述的结合，理论方法与经验方法的结合，精确方法与近似方法的结合，科学理性与艺术直觉的结合，等等。这些结合是系统论方法之精髓所在。

—《系统科学》，许国志

基本概念与方法

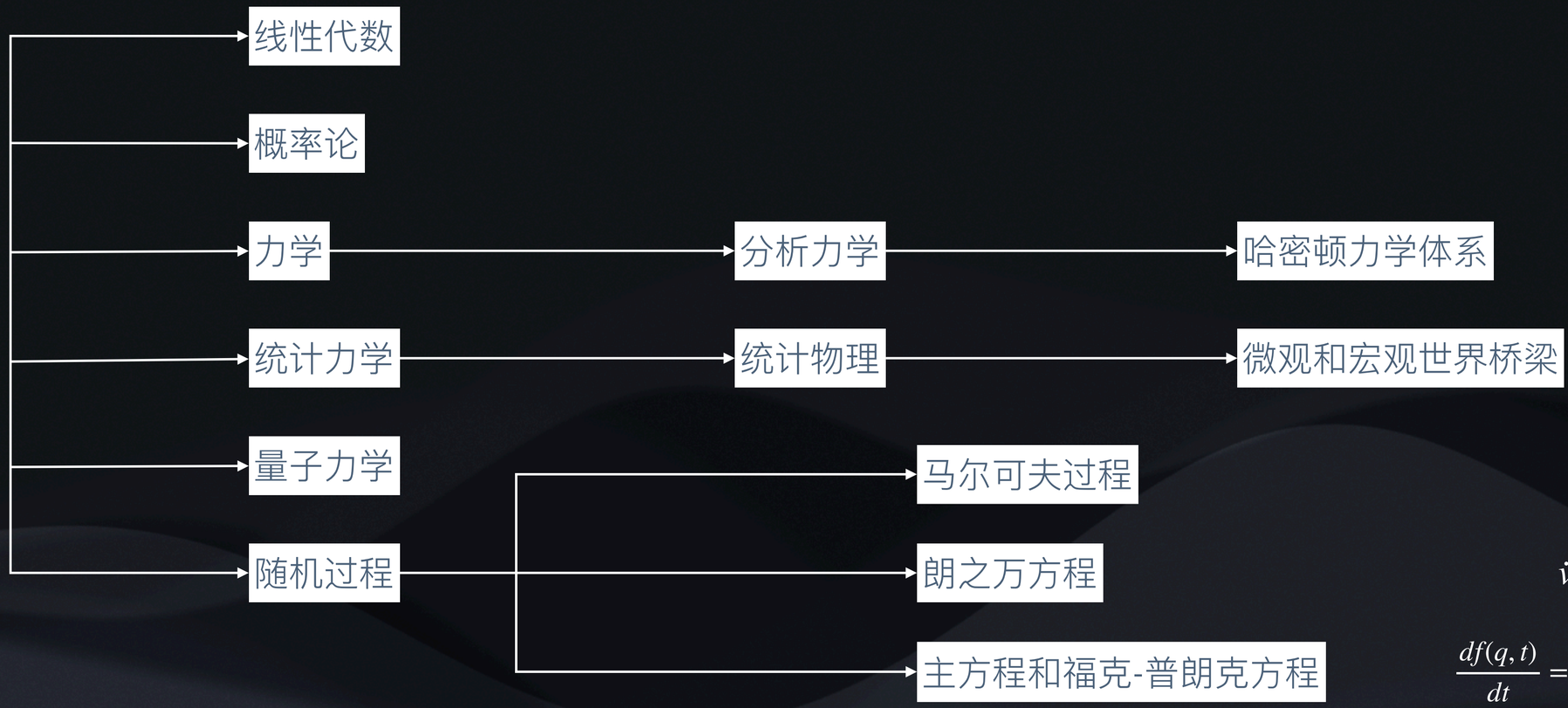

系统
如果一个对象集合中至少有两个可以区分的对象，所有对象按照可以辨认的特有方式相互联系在一起，就称该集合为一个系统
结构
组分及组分之间关联方式（系统把其元素整合为统-整体的模式）的总和
层次
涌现性—质的提升，量变质变
环境
一个系统之外的一切与它相关联的事物构成的集合
行为
系统相对于它的环境所表现出来的任何变化
功能
系统行为所引起的，有利环境中某些事物乃至整个环境存续与发展的作用
状态
系统的那些可以观察和识别的状况、态势、特征等
演化
系统的结构、状念、特性、行为、功能等随着时间的推移而发生的变化
过程
凡演化都是过程——重要分类（是否时间可逆）



任何系统都有定性特性和定量特性两方面，定性特性决定定量特性，定量特性表现定性特性：

1. 随机性： BS Model，随机过程
2. 基本面： 财务报告
3. 跳跃现象：泊松过程
4. 时间序列： 系统演化

系统科学的数学物理基础

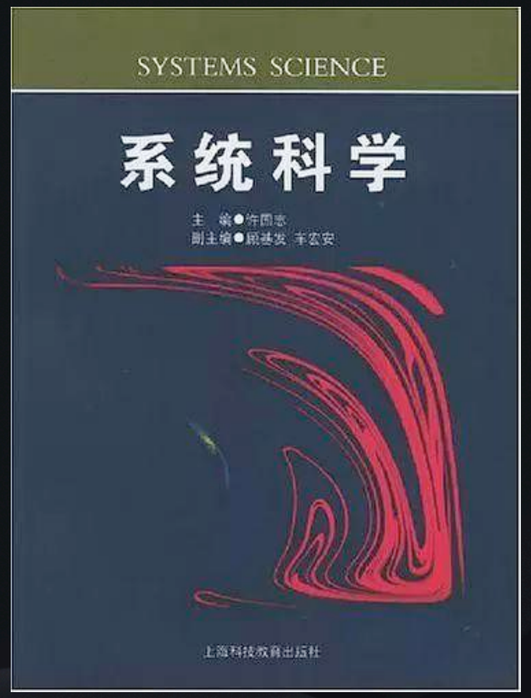


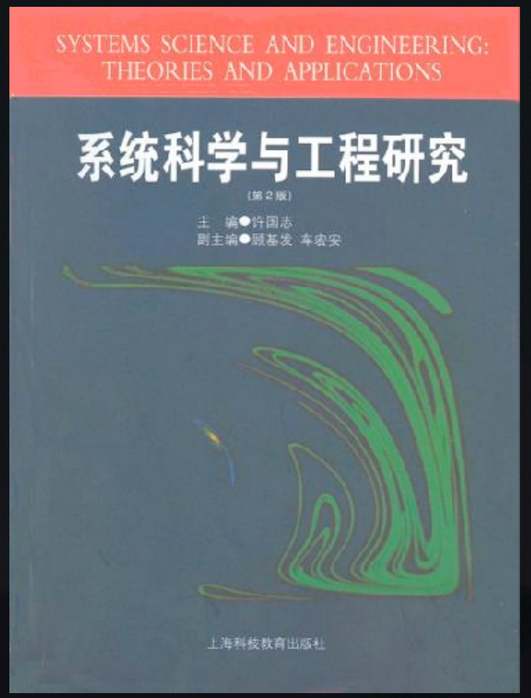
$$\dot{v} = -\frac{\gamma}{m}v + \frac{1}{m}\xi_t$$

$$\frac{df(q,t)}{dt} = \frac{d}{dq}(\gamma q f(q,t)) + \frac{D}{2}\frac{d^2}{dq^2}f(q,t)$$

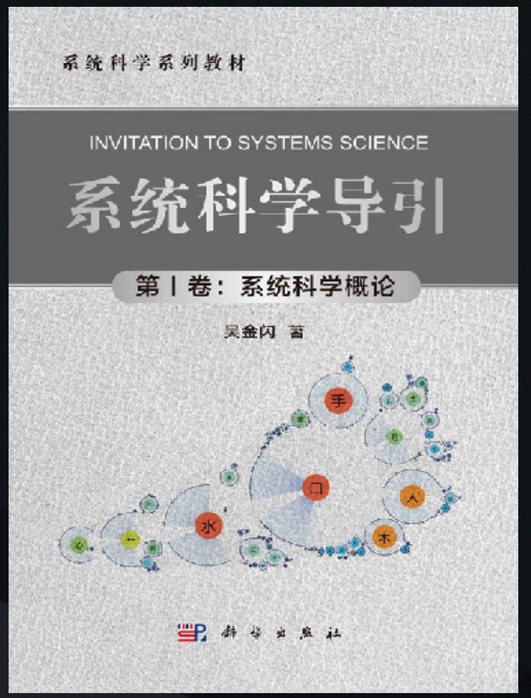
$$\frac{df(q,t)}{dt} = \frac{D}{2}\frac{d^2}{dq^2}f(q,t)$$

参考教材









1. 系统工程理论与实践， https://sysengi.cjoe.ac.cn/CN/home

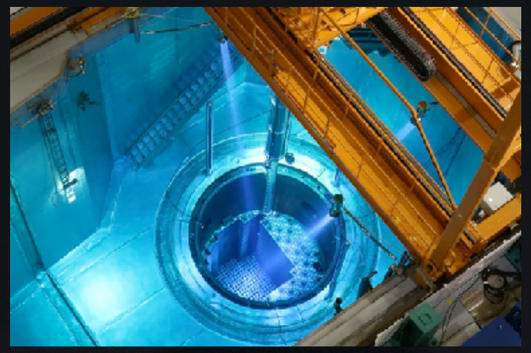
2. 集智俱乐部， https://swarma.org/

实例分享

核电站

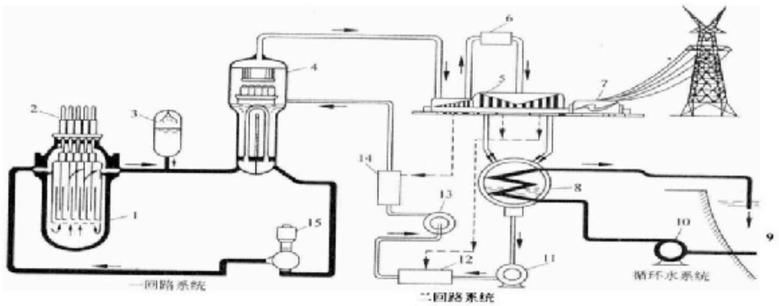






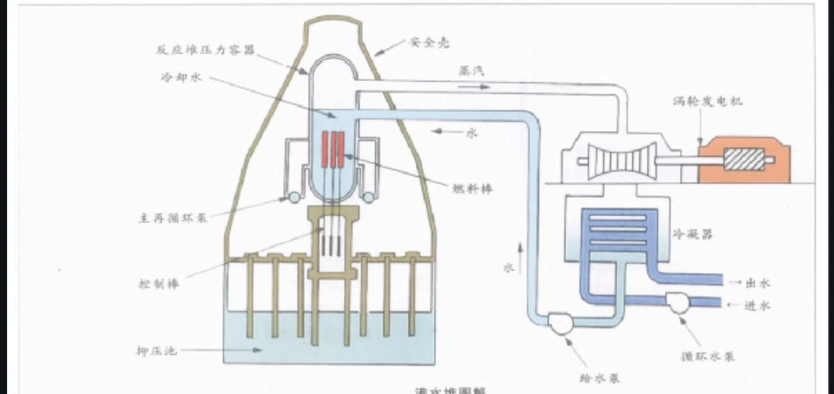


图表9：压水堆核电站工作原理示意图

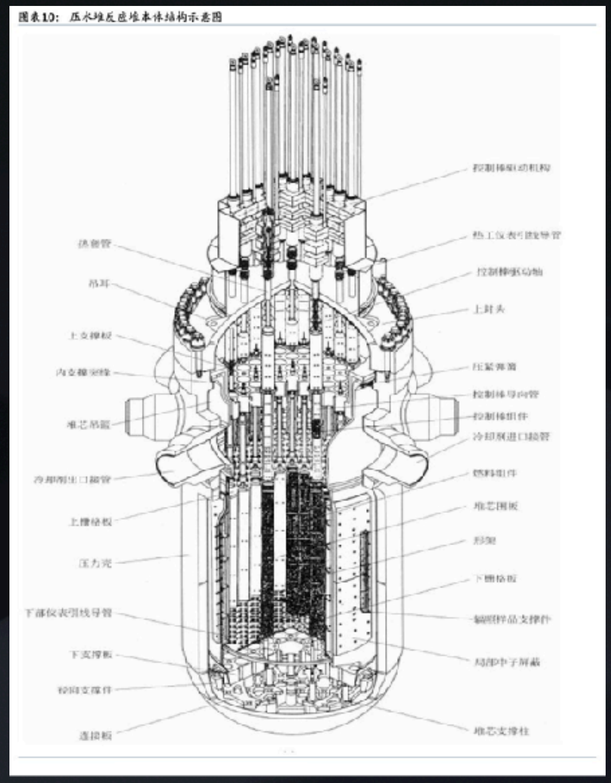


1—反应堆压力容器；2—控制棒传动机构；3—稳压器；4—蒸汽发生器；5—汽轮机；6—汽水分离再热器；7—发电机；8—凝汽器；9—循环水水源；10—循环水泵；11—凝结水泵；12—低压加热器；13—给水泵；14—高压加热器；15—反应堆冷却剂泵

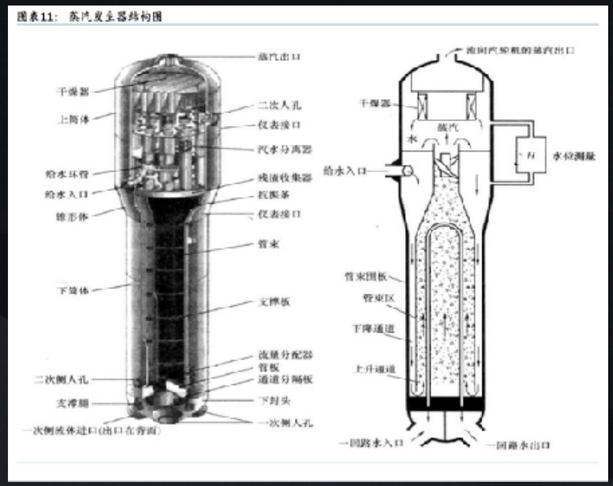
图表29：沸水堆核电站工作原理示意图



图表10：压水堆反应堆本体结构示意图



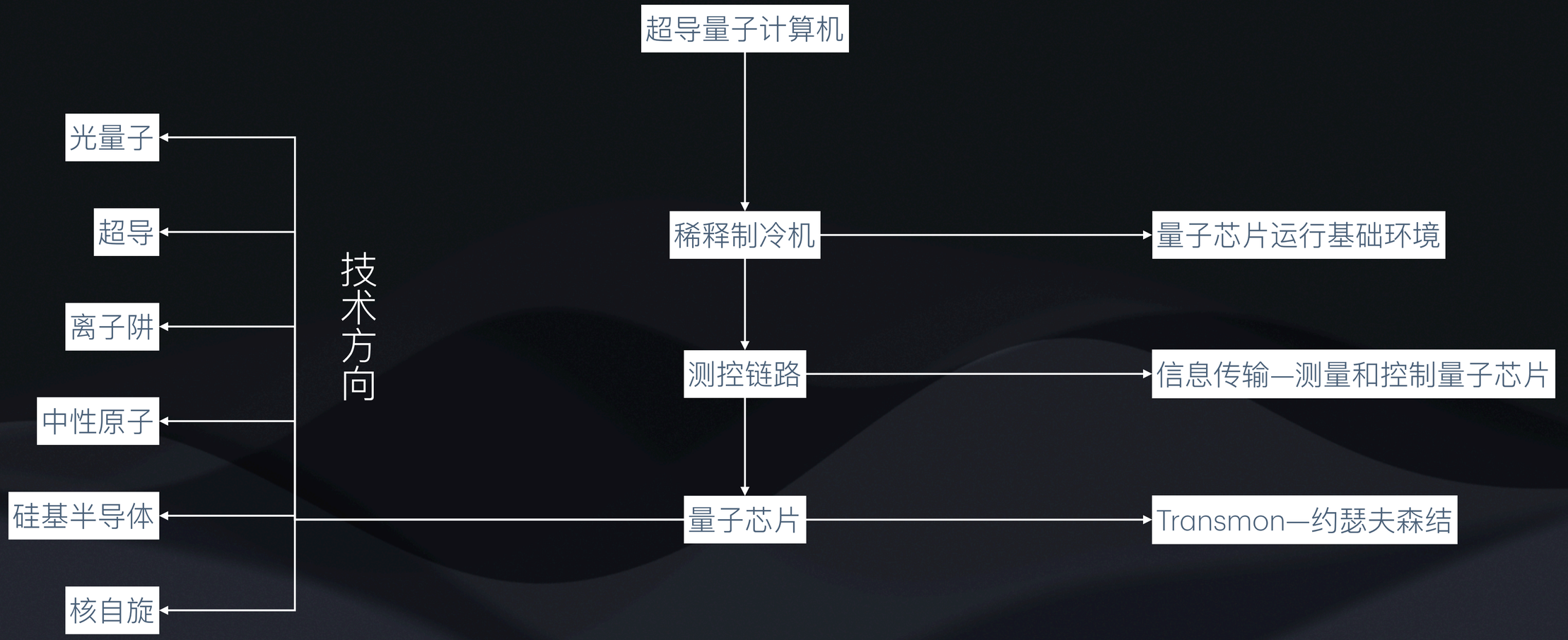
图表11：蒸汽发生器结构图



中国核电网—https://www.cnnpn.cn/channel/6.html?&page=2

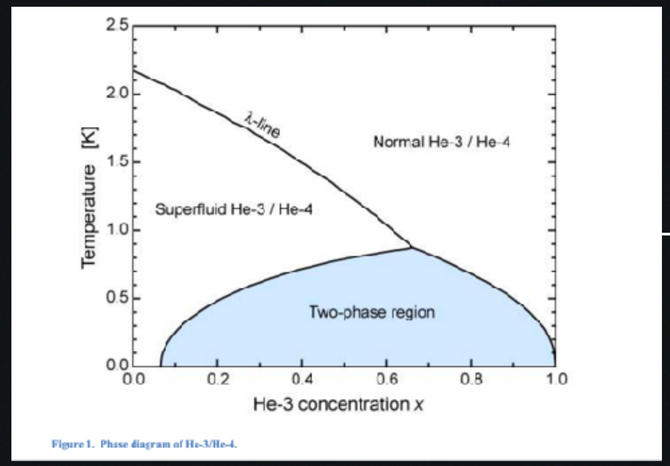
超导量子计算机

系统架构

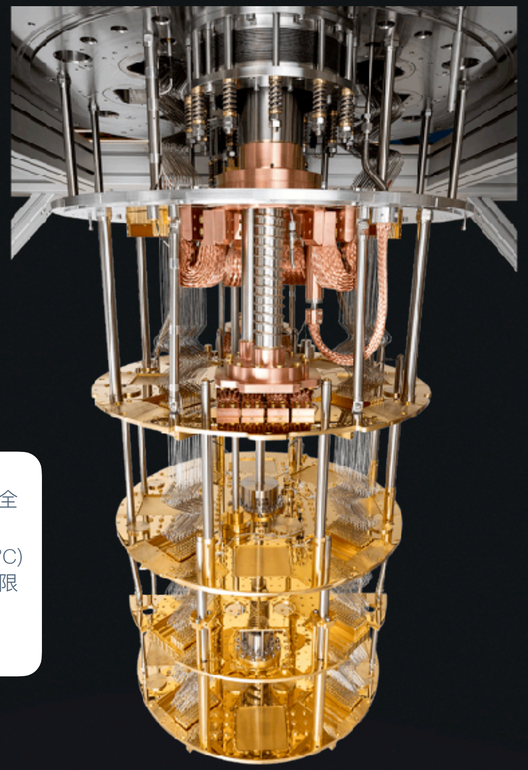


稀释制冷机

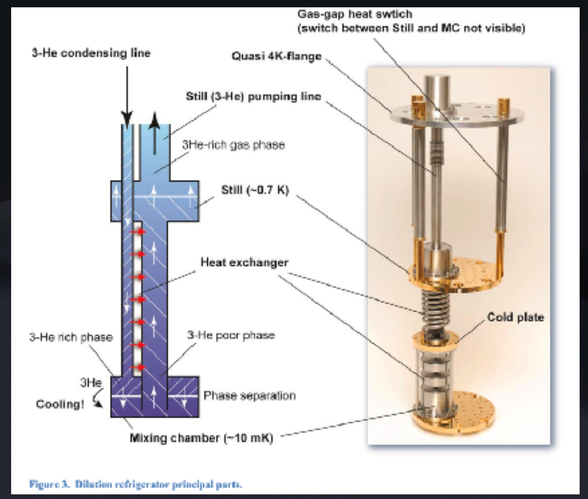
原理—热力学、低温物理

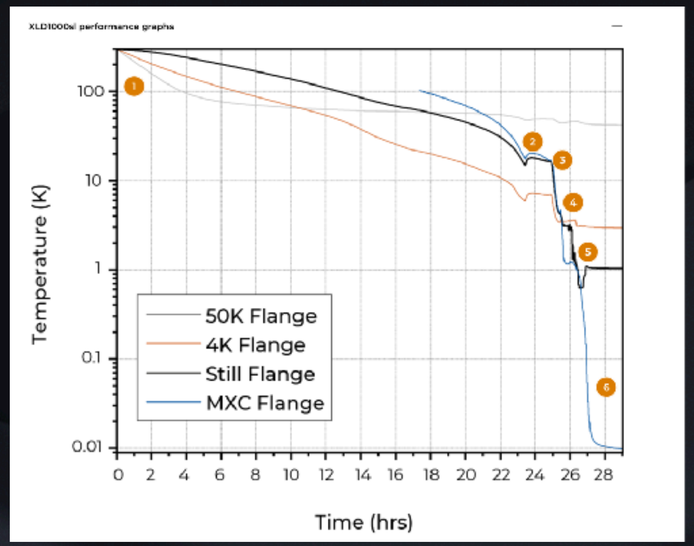


超流体：
一种物质状态，特点是完全缺乏黏性；
1. He-4 — <2.17K(−270.98°C)全无粘性、零熵度，和无限大的热传导率
2. He-3 — <2.6mk









热力学三大定律

1. 能量守恒与转化
2. 热量不可能自动地由低温物体向高温物体传递，不可逆性—熵增加原理
3. 绝对零度不可能达到
4. 范德瓦尔斯方程：

$$(P + \frac{a}{V^2})(V - b) = RT$$

1mol气体压强P、体积V和温度T之间关系，R是普适气体常数，a和b是实际气体参数。

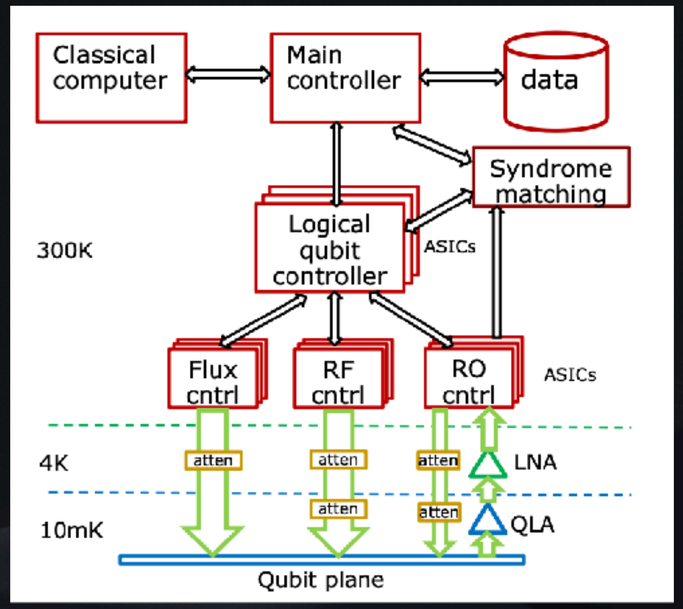
——量子力学，顾樵

绝对零度-273.15°C = 0K

https://bluefors.com/

测控链路

微波工程、电子电路、低温电子学



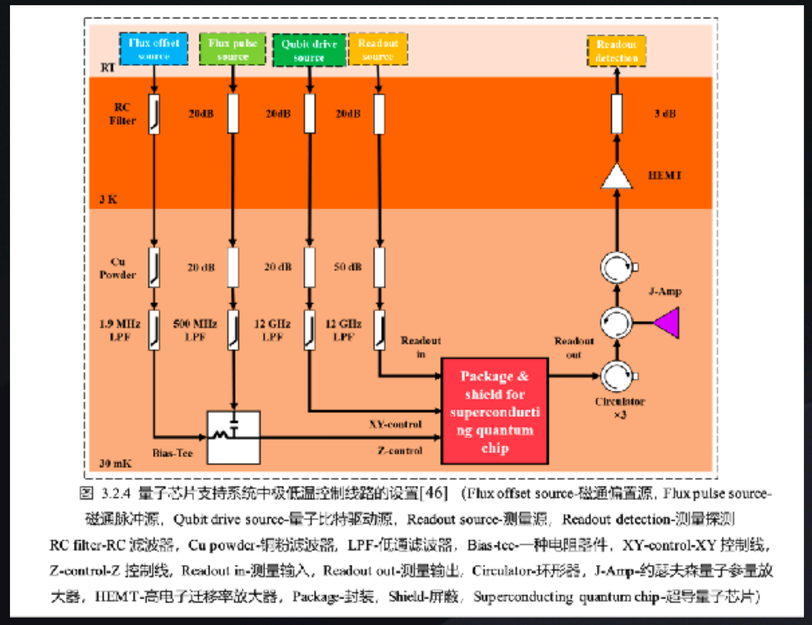
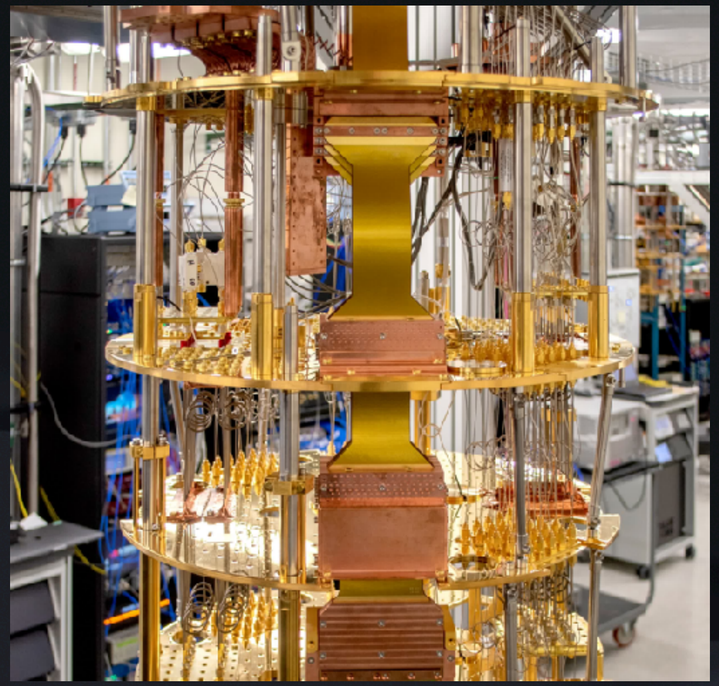


图 3.2.4 量子芯片支持系统中级低温控制线路的设置[46]（Flux offset source-磁通偏置源，Flux pulse source-磁通脉冲源，Qubit drive source-量子比特驱动源，Readout source-测量源，Readout detection-测量探测 RC filter-RC 滤波器，Cu powder-铜粉滤波器，LPF-低通滤波器，Bias-tee-一种电阻器件，XY-control-XY 控制线，Z-control-Z 控制线，Readout in-测量输入，Readout out-测量输出，Circulator-环形器，J-Amp-约瑟夫森量子参量放大器，HEMT-高电子迁移率放大器，Package-封装，Shield-屏蔽，Superconducting quantum chip-超导量子芯片）



超导量子芯片

基础数学

希尔伯特空间：无限维完备的欧几里得空间叫做希尔伯特空间
满足以下公里任意性质的元素f,g，...的全体H叫做希尔伯特空间

1. H是欧几里得空间（即在其中具有给定内积的线性空间）；
2. 空间H在度量ρ(f,g) = ||f - g||的意义下是完备的；
3. 空间H无限维的，即在H中对于任何n皆可找到n个线性无关的元素；
4. H是可分的，即在H中存在可数处处稠密集。

欧几里得空间：实线性空间R中，每一对元素$x, y \in R$定义的实函数(x, y)，如果满足以下条件：

1. $(x, y) = (y, x)$
2. $(x_1 + x_2, y) = (x_1, y) + (x_2, y)$
3. $(\lambda x, y) = \lambda(x, y)$
4. $(x, x) \geq 0$, 并且$(x, x) = 0$ 仅当$x = 0$

在其中具有确定内积的线性空间叫做欧几里得空间。引进范数，

$$||x|| = \sqrt{(x, x)}$$

这时所有范数公里是满足的。

—函数论与泛函分析初步，A. H. 科尔莫戈洛夫、C. B. 佛名

复数（二元数）是成对存在，伴随其对偶形式：

$$z = a \pm ib$$

欧拉公式：

$$e^{ix} = cos(x) + isin(x)$$

$$z_0 = e^{i\theta}$$

$$z_0 = \begin{pmatrix} cos(\theta) & -sin(\theta) \\ sin(\theta) & cos(\theta) \end{pmatrix}$$

$$z_0 z = e^{i\theta} r e^{i\varphi} = r e^{i(\theta+\varphi)}$$

内积：

$$(|v\rangle, |w\rangle) = \langle v|w\rangle = \sum_i y_i^* z_i = \begin{bmatrix} y_1^* \cdots y_n^* \end{bmatrix} \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix}$$

张量积：

$$\begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix} \otimes \begin{bmatrix} b_1 \\ \vdots \\ b_k \end{bmatrix} = \begin{bmatrix} a_1 b_1 \\ \vdots \\ a_1 b_k \\ \vdots \\ a_m b_1 \\ \vdots \\ a_m b_k \end{bmatrix}$$

态矢量由抽象希尔伯特空间子空间（有限维）描述：

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} and\ |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\langle 0| = [1 \quad 0]\ and\ \langle 1| = [0 \quad 1]$$

〈 |**左矢（bra）**是其对偶空间的矢量，| 〉是**右矢（ket）**。

超导量子芯片

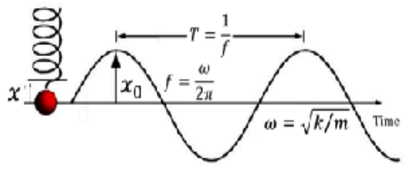
简单思路

- 二进制：010000100010（机器码）
- 传统计算机：0或1代表一个信息比特，由晶体管单独体现0或1，“开”或“关”
- 量子计算机：构建二能级体系包含（0，1）一个整体信息比特，此为一个量子比特
- 超导量子芯片由约瑟夫森结搭建，一个非线性电感元件

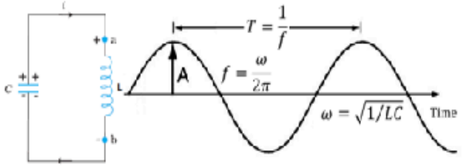
超导量子芯片

约瑟夫森结

简谐振动



弹簧振子



LC 回路

- 简谐振动可以用含时的正弦和余弦函数来表示

$$x(t)=x_0\sin(\omega t+\varphi)\sim x_0\sin(\omega t)$$

$$p(t)=m\frac{dx}{dt}=p_0\cos(\omega t+\varphi)\sim p_0\cos(\omega t)$$

$$E=\frac{p^2}{2m}+\frac{1}{2}m\omega^2x^2=\frac{1}{2m}p_0^2\cos^2(\omega t)+\frac{m\omega^2}{2}x_0^2\sin^2(\omega t)$$

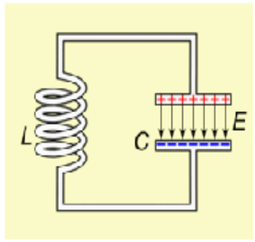
- LC回路

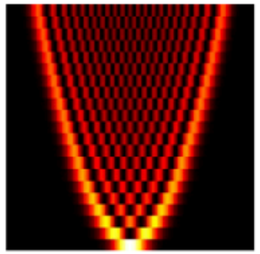
$$I(t)=I_0\sin(\omega t+\varphi)\sim I_0\sin(\omega t)$$

$$V(t)=L\frac{dI}{dt}=V_0\cos(\omega t+\varphi)\sim V_0\cos(\omega t)$$

$$E=\frac{1}{2}LI^2+\frac{1}{2}CV^2=\frac{LI_0^2}{2}\sin^2(\omega t)+\frac{CV_0^2}{2}\cos^2(\omega t)$$

量子LC回路





$$E=\frac{1}{2}LI^2+\frac{1}{2}CV^2=\frac{\Phi^2}{2L}+\frac{Q^2}{2C}$$

- 因此我们可以写下LC回路的哈密顿量

$$\hat{H}=\frac{\hat{\Phi}^2}{2L}+\frac{\hat{Q}^2}{2C}$$

- 对照简谐振子，引入产生消灭算符

$$\hat{a}^\dagger=-i\frac{1}{\sqrt{2C\hbar\omega}}\hat{Q}+\frac{1}{\sqrt{2L\hbar\omega}}\hat{\Phi}\qquad\hat{a}=i\frac{1}{\sqrt{2C\hbar\omega}}\hat{Q}+\frac{1}{\sqrt{2L\hbar\omega}}\hat{\Phi}$$

- 所以

$$\hat{H}=\hbar\omega\left\{\hat{a}^\dagger\hat{a}+\frac{1}{2}\right\}$$

- 算符之间满足如下对易关系

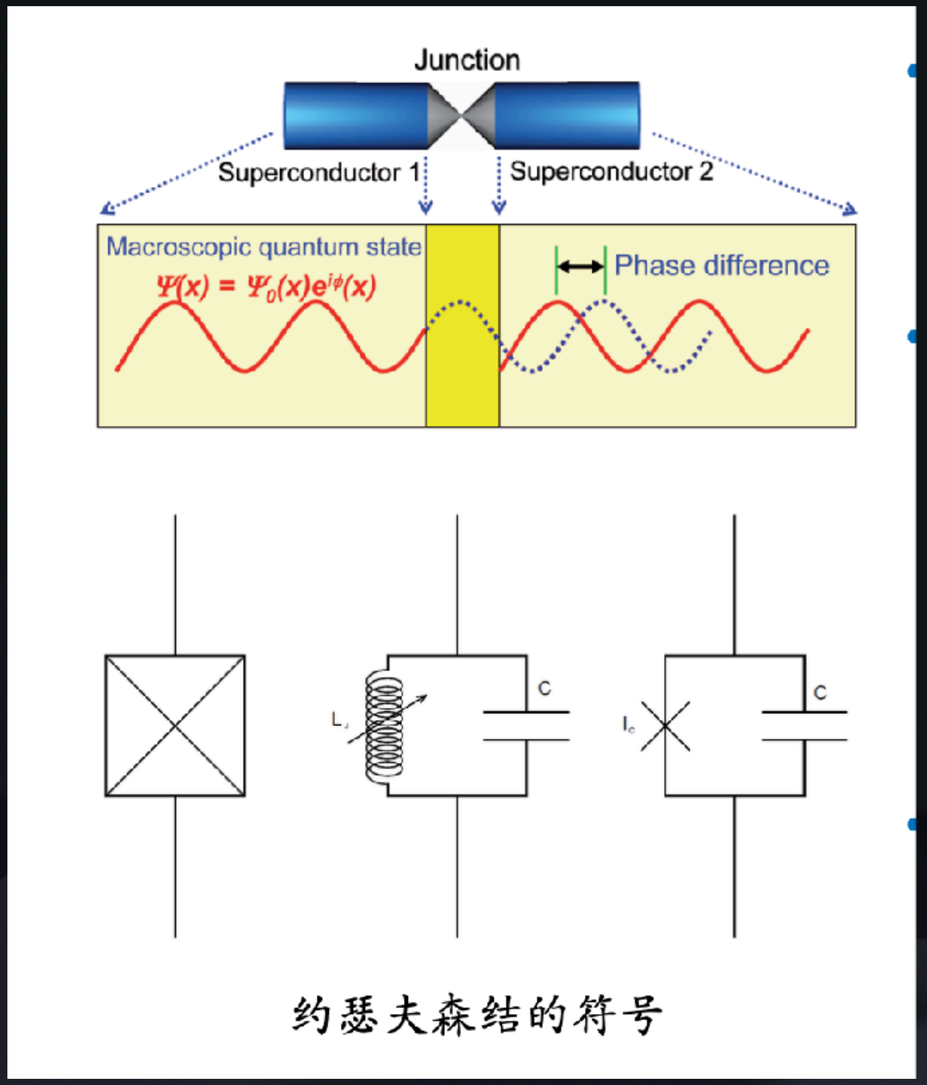
$$\left[\hat{\Phi},\hat{Q}\right]=i\hbar\qquad\left[\hat{a},\hat{a}^\dagger\right]=1$$

量子前沿展望/金潮渊：量子信息基础/集智俱乐部—https://campus.swarma.org/course/3099

超导量子芯片

约瑟夫森结

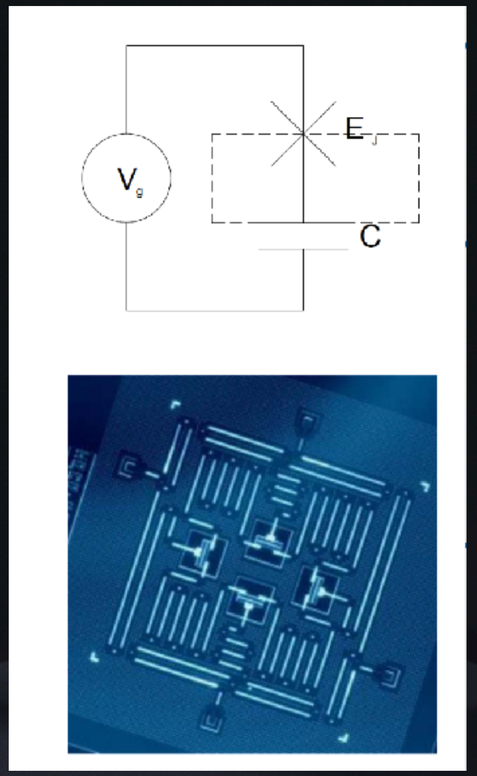
- 约瑟夫森效应是宏观量子效应的体现，以英国物理学家布莱恩·约瑟夫森（Brian D.Josephson)命名
- 在量子力学中，非超导状态的电子可以借由量子遂穿效应流过绝缘层。约瑟夫森首次预测超导状态下库伯对的遂穿现象，因此获得1973年诺贝尔物理学奖。
- 约瑟夫森结由间隔一段绝缘势垒的两段超导体构成，势垒的宽度在纳米量级。约瑟夫森结非线性主要体现在电子形成的库伯对（Copper pair）只能成对地通过结区。

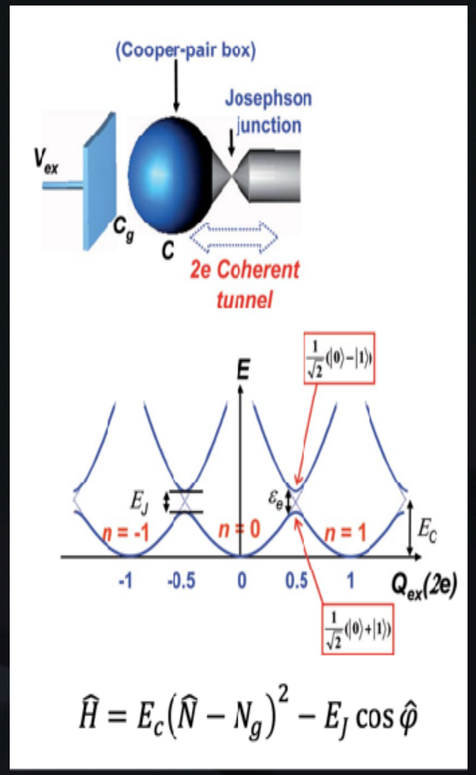


约瑟夫森结的符号

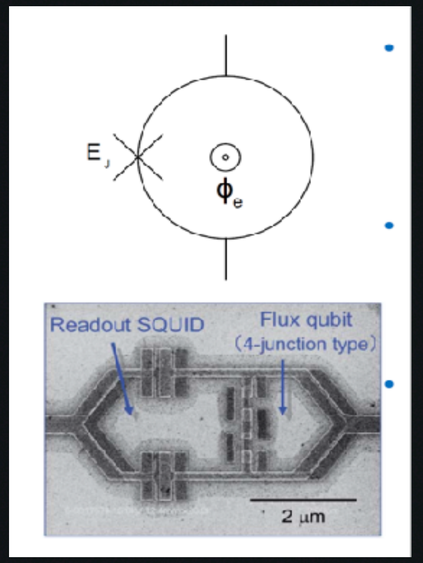
超导量子芯片

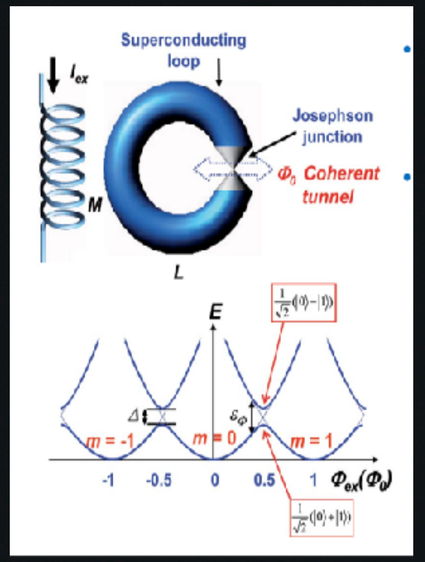
约瑟夫森结



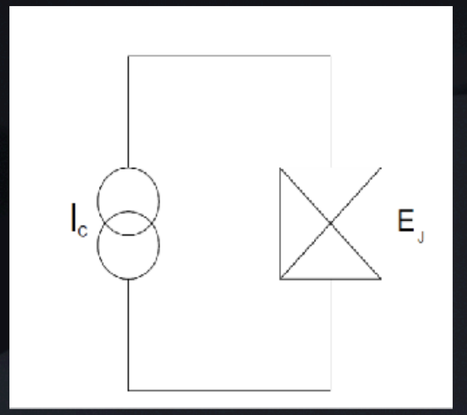


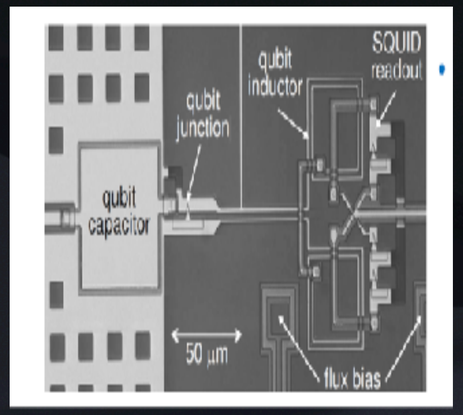
超导电荷量子比特





超导磁通量子比特



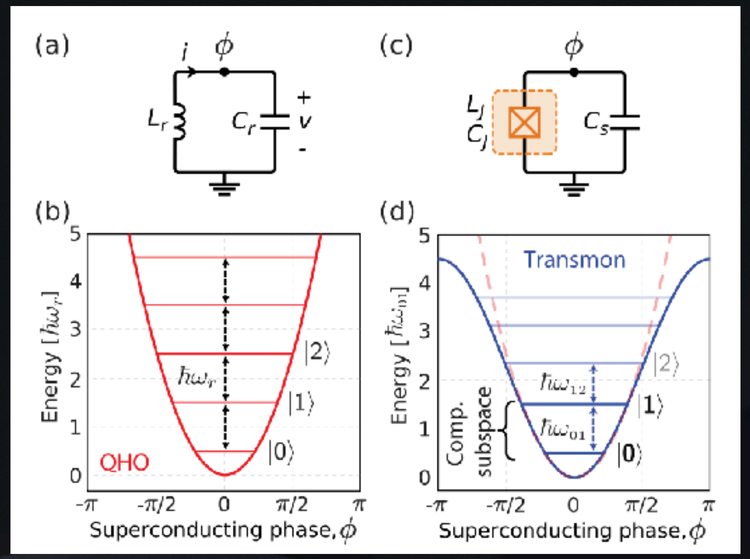


超导相位量子比特

电流源和约瑟夫森

超导量子芯片

约瑟夫森结



能级劈裂

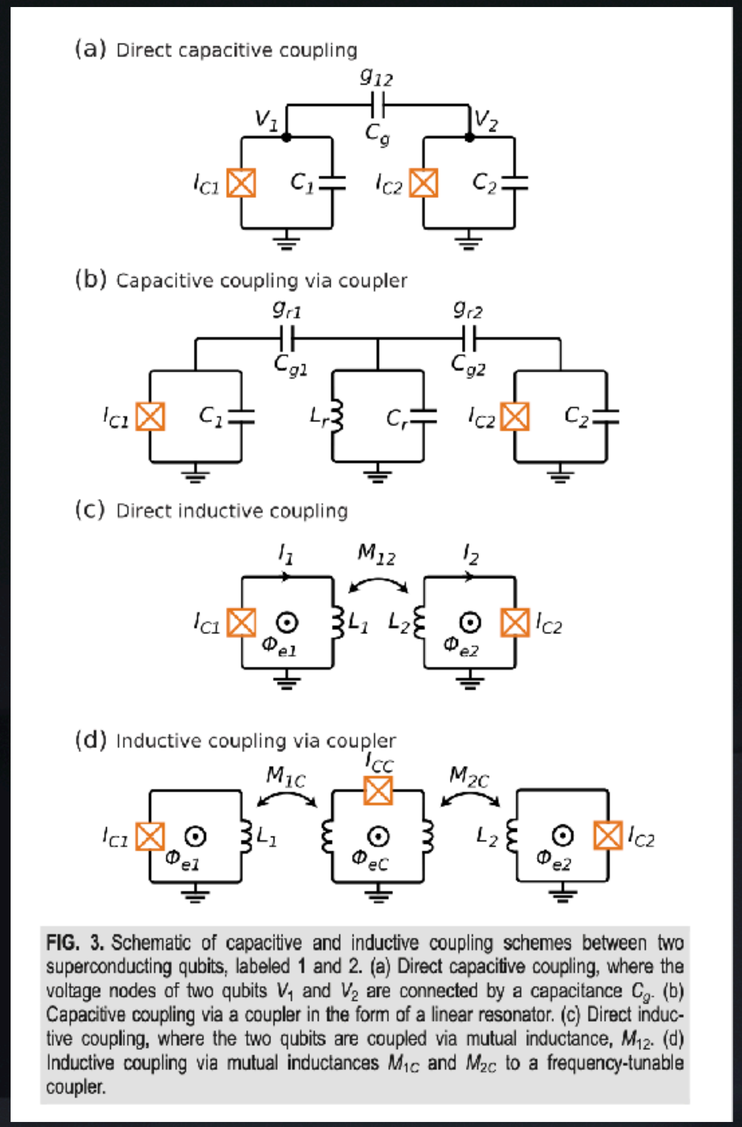
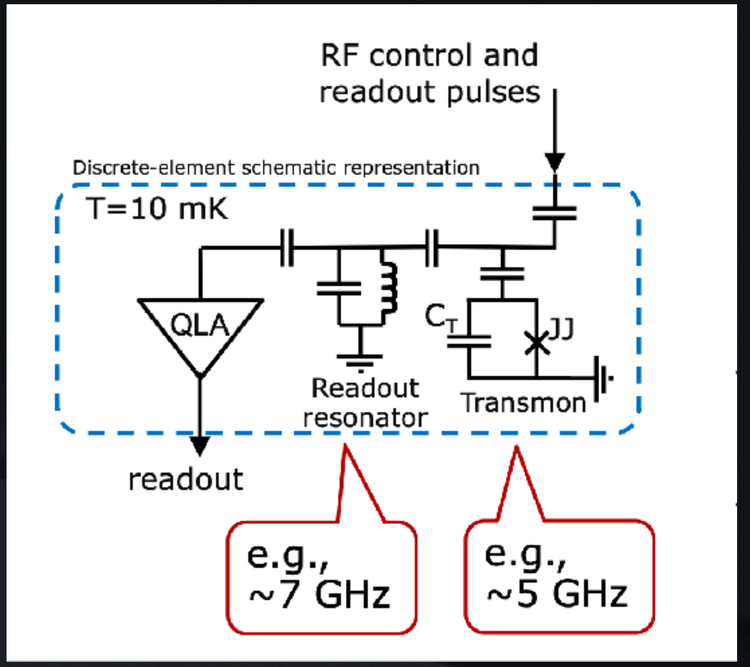


FIG. 3. Schematic of capacitive and inductive coupling schemes between two superconducting qubits, labeled 1 and 2. (a) Direct capacitive coupling, where the voltage nodes of two qubits V_1 and V_2 are connected by a capacitance C_g. (b) Capacitive coupling via a coupler in the form of a linear resonator. (c) Direct inductive coupling, where the two qubits are coupled via mutual inductance, M_{12}. (d) Inductive coupling via mutual inductances M_{1C} and M_{2C} to a frequency-tunable coupler.

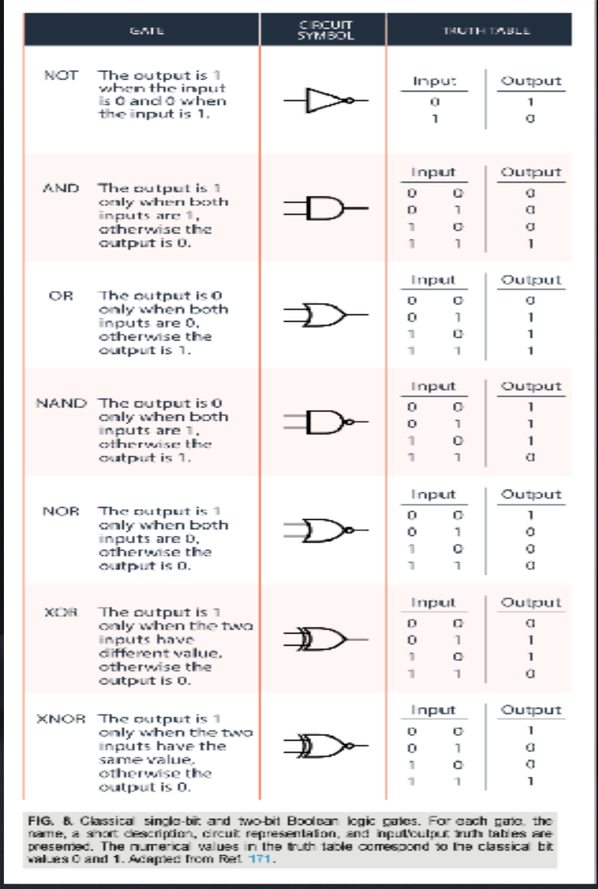
耦合量子比特



读出电路

量子计算

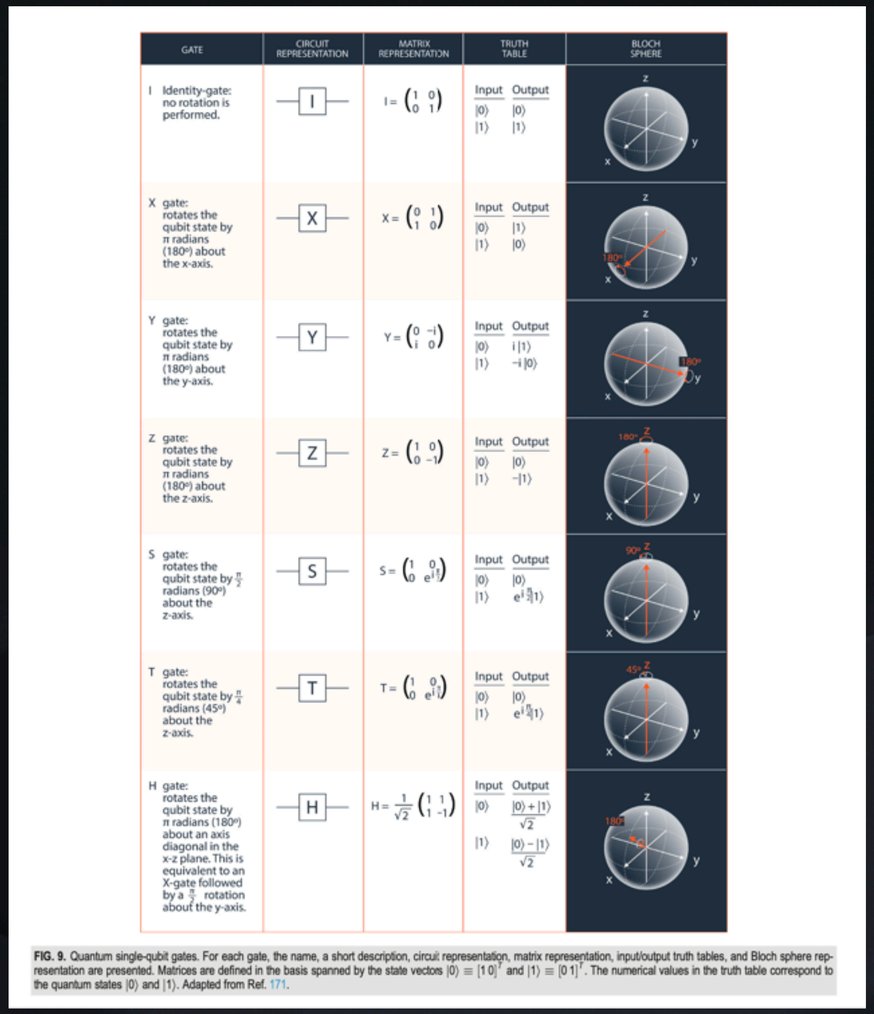
布尔逻辑门



GATE	CIRCUIT SYMBOL	TRUTH TABLE
NOT The output is 1 when the input is 0 and 0 when the input is 1.		Input: 0, 1 — Output: 1, 0
AND The output is 1 only when both inputs are 1, otherwise the output is 0.		Input: 0 0, 0 1, 1 0, 1 1 — Output: 0, 0, 0, 1
OR The output is 0 only when both inputs are 0, otherwise the output is 1.		Input: 0 0, 0 1, 1 0, 1 1 — Output: 0, 1, 1, 1
NAND The output is 0 only when both inputs are 1, otherwise the output is 1.		Input: 0 0, 0 1, 1 0, 1 1 — Output: 1, 1, 1, 0
NOR The output is 1 only when both inputs are 0, otherwise the output is 0.		Input: 0 0, 0 1, 1 0, 1 1 — Output: 1, 0, 0, 0
XOR The output is 1 only when the two inputs have different value, otherwise the output is 0.		Input: 0 0, 0 1, 1 0, 1 1 — Output: 0, 1, 1, 0
XNOR The output is 1 only when the two inputs have the same value, otherwise the output is 0.		Input: 0 0, 0 1, 1 0, 1 1 — Output: 1, 0, 0, 1

FIG. 8. Classical single-bit and two-bit Boolean logic gates. For each gate, the name, a short description, circuit representation, and input/output truth tables are presented. The numerical values in the truth table correspond to the classical bit values 0 and 1. Adapted from Ref. 171.

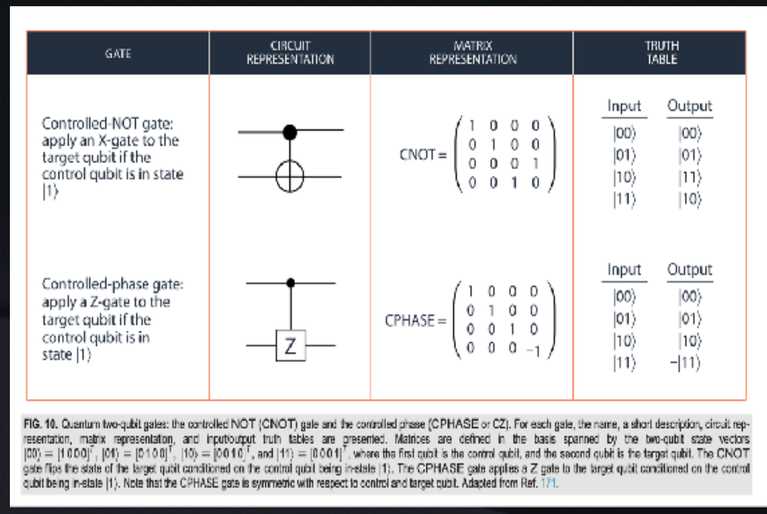
经典计算机



GATE	CIRCUIT REPRESENTATION	MATRIX REPRESENTATION	TRUTH TABLE	BLOCH SPHERE
I Identity-gate: no rotation is performed.	I	$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$	Input: $\|0\rangle$, $\|1\rangle$ — Output: $\|0\rangle$, $\|1\rangle$	
X gate: rotates the qubit state by π radians (180°) about the x-axis.	X	$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	Input: $\|0\rangle$, $\|1\rangle$ — Output: $\|1\rangle$, $\|0\rangle$	
Y gate: rotates the qubit state by π radians (180°) about the y-axis.	Y	$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	Input: $\|0\rangle$, $\|1\rangle$ — Output: $i\|1\rangle$, $-i\|0\rangle$	
Z gate: rotates the qubit state by π radians (180°) about the z-axis.	Z	$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	Input: $\|0\rangle$, $\|1\rangle$ — Output: $\|0\rangle$, $-\|1\rangle$	
S gate: rotates the qubit state by $\frac{\pi}{2}$ radians (90°) about the z-axis.	S	$S = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{2}} \end{pmatrix}$	Input: $\|0\rangle$, $\|1\rangle$ — Output: $\|0\rangle$, $e^{i\frac{\pi}{2}}\|1\rangle$	
T gate: rotates the qubit state by $\frac{\pi}{4}$ radians (45°) about the z-axis.	T	$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\frac{\pi}{4}} \end{pmatrix}$	Input: $\|0\rangle$, $\|1\rangle$ — Output: $\|0\rangle$, $e^{i\frac{\pi}{4}}\|1\rangle$	
H gate: rotates the qubit state by π radians (180°) about an axis diagonal in the x-z plane. This is equivalent to an X-gate followed by a $\frac{\pi}{2}$ rotation about the y-axis.	H	$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	Input: $\|0\rangle$, $\|1\rangle$ — Output: $\frac{\|0\rangle + \|1\rangle}{\sqrt{2}}$, $\frac{\|0\rangle - \|1\rangle}{\sqrt{2}}$	

FIG. 9. Quantum single-qubit gates. For each gate, the name, a short description, circuit representation, matrix representation, input/output truth tables, and Bloch sphere representation are presented. Matrices are defined in the basis spanned by the state vectors $|0\rangle \equiv [1\,0]^T$ and $|1\rangle \equiv [0\,1]^T$. The numerical values in the truth table correspond to the quantum states $|0\rangle$ and $|1\rangle$. Adapted from Ref. 171.

单比特



GATE	CIRCUIT REPRESENTATION	MATRIX REPRESENTATION	TRUTH TABLE
Controlled-NOT gate: apply an X-gate to the target qubit if the control qubit is in state $\|1\rangle$		$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$	Input: $\|00\rangle$, $\|01\rangle$, $\|10\rangle$, $\|11\rangle$ — Output: $\|00\rangle$, $\|01\rangle$, $\|11\rangle$, $\|10\rangle$
Controlled-phase gate: apply a Z-gate to the target qubit if the control qubit is in state $\|1\rangle$	Z	$\text{CPHASE} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$	Input: $\|00\rangle$, $\|01\rangle$, $\|10\rangle$, $\|11\rangle$ — Output: $\|00\rangle$, $\|01\rangle$, $\|10\rangle$, $-\|11\rangle$

FIG. 10. Quantum two-qubit gates: the controlled NOT (CNOT) gate and the controlled phase (CPHASE or CZ). For each gate, the name, a short description, circuit representation, matrix representation, and input/output truth tables are presented. Matrices are defined in the basis spanned by the two-qubit state vectors $|00\rangle = [1\,0\,0\,0]^T$, $|01\rangle = [0\,1\,0\,0]^T$, $|10\rangle = [0\,0\,1\,0]^T$, and $|11\rangle = [0\,0\,0\,1]^T$, where the first qubit is the control qubit, and the second qubit is the target qubit. The CNOT gate flips the state of the target qubit conditioned on the control qubit being in state $|1\rangle$. The CPHASE gate applies a Z gate to the target qubit conditioned on the control qubit being in state $|1\rangle$. Note that the CPHASE gate is symmetric with respect to control and target qubit. Adapted from Ref. 171.

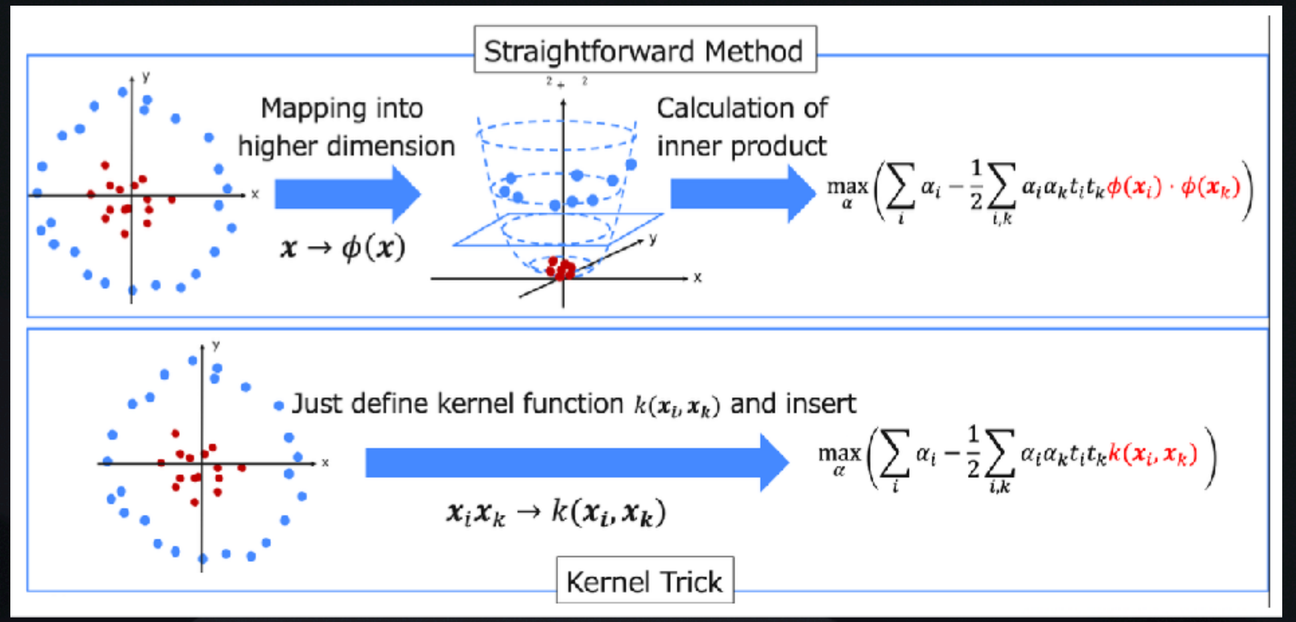
两比特

量子计算

量化金融应用

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- 投资组合优化
- 机器学习、深度学习
- 多维偏微分方程求解（深度学习）
- 期权组合定价
- 蒙特卡洛模拟

Quantum Support Vector Method（支持向量机）



$$k(x_i, x) := \phi(x_i)^\dagger \phi(x) = tr[\rho(x)\rho(y)] \equiv |\langle \psi(x) | \psi(y) \rangle|^2 \equiv |\langle 0 | U(x)^\dagger U(y) | 0 \rangle|^2$$

—Quantum Complementarity and Potential for Advantage in Machine Learning，CQF Institute

IBM Qiskit Finance，https://qiskit-community.github.io/qiskit-finance/tutorials/index.html

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- 数学物理方法，R.柯朗, D.希尔伯特
- A quantum engineer's guide to superconducting qubits
- SC3.Superconducting.Qubits.Principles.ControlReadout.Architectures.and.Cryoelectronic.Solutions
- Principles of Superconducting Quantum Computers, Daniel D. Stancil, Gregory T. Byrd
- Quantum Economics and Finance An Applied Mathematics Introduction, David Orrell
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